Summary in Graph

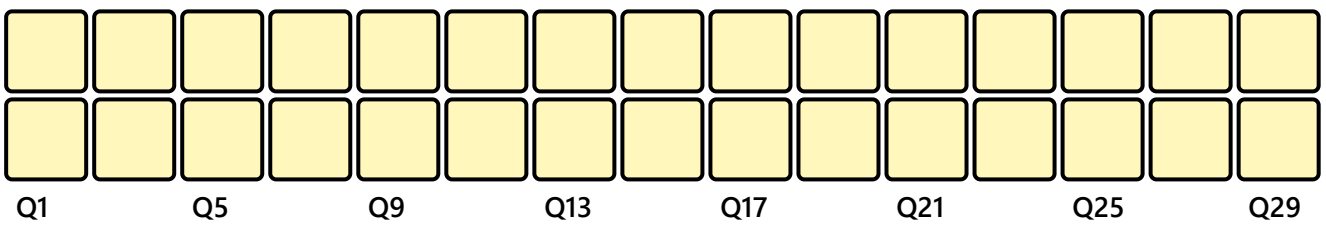
# Exam Summary (GATE Overflow CS Test Series 2026 | Data Structures | Test 1)

| | |
|---|---|
| Qs. Attempted: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) |
| Correct Marks: | 0 (0 + 0) |
| Penalty Marks: | 0 (0 + 0) |
| Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 30 (10 + 20) |
| Total Marks: | 50 (10 + 40) |
| Exam Duration: | 90 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained



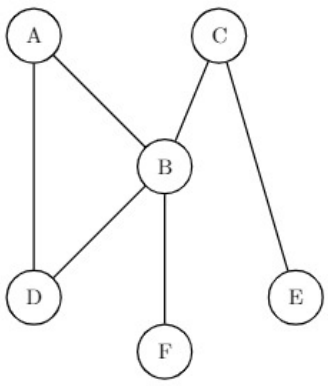

* Average Stats based on first-attempt stats of Top 10% (6) out of 56 users

EXAM RESPONSE EXAM STATS FEEDBACK

## Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

The cut vertices in the given graph are




A. E and F
B. A, C and D
C. C, E and F
D. B and C

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Kevin is asked to write a single program to print the inorder, preorder, postorder and level order traversal of a binary tree having $n$ nodes. The best possible time complexity of this function is _____

A. $\Theta(n^3)$
B. $\Theta(n)$
C. $\Theta(n^2)$
D. $\Theta(n \log n)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

What is the value of the postfix expression $8\ 7\ 2\ 4\ +\ -\ *$?:

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The data structures most suitable to do an inorder and level order traversals of a binary tree respectively are

A. Stack and Queue
B. Queue and Stack
C. Stack and Stack
D. Queue and Queue

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The prefix form of $A/B - (C * D * E)$ is?

A. $-/**ABCDE$
B. $-/ABCD**DE$
C. $-/AB**CDE$

D. $-A/BC*DE*$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Which of the following is not a valid sequence of key traversals while searching the key $44$ in a binary search tree?

A. $11, 88, 22, 77, 33, 66, 44$
B. $11, 22, 33, 44$
C. $88, 11, 22, 77, 55, 44$
D. $66, 11, 22, 77, 33, 44$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

For a given simple graph $G$ having $v$ vertices and $e$ edges which is connected and has no cycles, which of the following statements is true?

A. $v = e$
B. $v = e + 1$
C. $v + 1 = e$
D. $v = e - 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

If the elements "A", "B", "C" and "D" in order are to be placed in a queue and stack and are deleted one element at a time (operations to the stack and queue being independent and deletion can happen in between insertions), which of the following is not a possible order for their removals from the queue and stack respectively?

A. ABCD, ABCD
B. DCBA, DCBA
C. ABCD, DCBA
D. All the above choices are possible

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

What is the location of parent node for any arbitrary node $i$ in an array implementation of a complete binary tree in C language?

A. floor$(i/2)$
B. floor$(i+1)/2$
C. floor$((i-1)/2)$
D. ceil$((i-1)/2)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #10** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Which of the following statements is TRUE regarding a doubly linked list?

A. It is possible to implement a doubly linked list with a single pointer at each node by storing XOR of addresses of previous and next nodes.
B. It is possible to implement a doubly linked list with a single pointer at each node by storing XOR of addresses of current node and next node
C. It is possible to implement a doubly linked list with a single pointer at each node by storing XOR of addresses of current node and previous node
D. We need minimum two pointers at each node for a doubly linked list implementation

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #11** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Let $L$ be a linked list of $n$ elements and $I$ be another linked list of $k$ node positions in $L$ $(n/4 < k < n/2)$ and elements in $I$ are not necessarily in sorted order. The best possible algorithm (use $O(1)$ extra space if required and assuming only comparison based sorting) to print the values in $L$ as per the node positions in $I$ will run in ____

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(n. k \log k)$
D. $\Theta(nk)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #12** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

The time complexity of the below code is

```
int isPalindrome ( char * string, int n )
{
    if (n <= 1) return 1;
    if (string[0] != string[n-1]) return 0;
    return isPalindrome(string + 1, n-2);
}
```

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta(\log n)$
D. $\Theta(n^2)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #13** Numerical Type | Award: 2 | Penalty: 0 | Data Structures

```
void fun(struct node* start)
{
    if(start == NULL)
        return;
    if(start->next != NULL )
        fun(start->next->next);
    printf("%d ", start->data);
}
```

The sum of the values printed by the above function when passed with the head to the following linked list is ____

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7$$

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #14

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following C function to traverse a linked list.

```
void traverse(struct node *root)
{
    while (root)
    {
        printf("%d ", root->data);
        root = root->next;
    }
}
```

Which of the following is $\mathbf{TRUE}$ about above function?

A. The function may crash when the passed linked list is empty
B. The function does not print the last node when the linked list is not empty
C. The function is implemented incorrectly because it changes the beginning of the linked list
D. The function is implemented correctly and print all the values of the linked list starting from root

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #15

Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Two balanced binary search trees are given with $m$ and $n$ elements respectively. They can be merged into another balanced binary search tree in $\Theta((m+n)^a(\log(m+n))^b)$ time. $2.a + 3b =$ ____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #16

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

The below code is for adding a new element to an array implementation of a min-heap. parent(i) returns the index of the parent node in the heap and size stores the number of elements in the heap.

```
void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}
void insert(int elem)
{
     size++;
    int i = size - 1;
    heap[i] = elem;
    while (i >= 0 && heap[parent(i)] > heap[i])
    {
     [FILL ME]
      i = parent(i);
    }
}
```

The correct replacement for [FILL ME] will be

A. `swap(&heap[i], &heap[parent(i)]);`

B. `swap(&heap[i], &heap[i+1]);`

C. `swap(&heap[i], &heap[2*i]);`

D. `swap(&heap[2*i], &heap[parent(i)]);`

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #17** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Given an array of elements $5, 7, 9, 1, 3, 10, 8, 4$, which of the following is not a possible array implementation of min-heap of these elements?

A. $1, 3, 4, 7, 5, 8, 9, 10$
B. $1, 4, 3, 8, 9, 5, 7, 10$
C. $1, 3, 4, 5, 8, 7, 9, 10$
D. All are min-heaps

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #18** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Let $f(n), g(n)$ and $h(n)$ be the time complexities of three programs. Let $f(n) = O(g(n))$ and $h(n) = O(g(n))$. Which of the following is necessarily FALSE?

A. $f(n) = O(h(n))$
B. $g(n) = O(f(n))$
C. $h(n) = \Omega(g(n))$
D. None of the above are necessarily FALSE

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #19** Numerical Type | Award: 2 | Penalty: 0 | Data Structures

The maximum number of leaf nodes in a tree with degree $5$ and height (starting from $1$) $4$ is _____

Your Answer: | Correct Answer: 125 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #20** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

What is the best time complexity (not the best case input but the best algorithm which will work for any input) for finding the lowest common ancestor for two keys $a$ and $b$ in a binary search tree of $n$ nodes and height $h$?

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $O(h)$
D. $\Omega(n)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 01sec | Discuss

**Q #21** Numerical Type | Award: 2 | Penalty: 0 | Data Structures

If the best time complexity (not the best case input but the best algorithm which will work for any input) for finding the shortest path between two keys $a$ and $b$ in a balanced binary search tree of $n$ nodes is denoted by $\Theta(n^a(\log n)^b)$, then $2^a \times 3^b =$____

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #22** Numerical Type | Award: 2 | Penalty: 0 | Data Structures

In a binary search tree the following key values (not necessarily in the order given) are encountered while searching for the key $29$.

$$3, 7, 13, 19, 37, 43, 51, 59.$$

The total number of possible orders in which the given keys of the binary search tree could have been visited is ____

Your Answer: | Correct Answer: 70 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #23** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

What will be the output of the C program?

```c
#include<stdio.h>
int main()
{
    int a[][4] = {0, 1, 2, 3, 4, 5, 6, 7};
    int (*ptr)[4] = a;
    printf("%d %d ", (*ptr)[0], (*ptr)[1]);
    ++ptr;
    printf("%d %d\n", (*ptr)[0], (*ptr)[1]);
    return 0;
}
```

A. $0\ 1\ 3\ 4$
B. $0\ 1\ 4\ 5$

C. 0 1 2 3
D. 0 1 6 7

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider the following representation of a ternary tree where $A : B, C, D$ denote that $B, C$ and $D$ are the child nodes of $A$.

- $1 : 2, 3, 4$
- $2 : 5, 6$
- $3 : 7$
- $4 : 8, 9, 10$

If the root of the above tree is passed to the following C function (assume the the child nodes are represented by child1, child2 and child3 and all NULL values are assigned properly), the return value will be ____

```
int foo(struct node *root)
{
    if(!root) return 0;
    return 1 + foo(root->child1) +
        foo(root->child2) +
        foo(root->child3);
}
```

Your Answer: | Correct Answer: 10 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Let $f(n), g(n)$ and $h(n)$ be the time complexities of three programs and let $f(n) = o(g(n))$ and $h(n) = \omega(g(n))$. Which of the following is VALID?

A. $f(n) = \Theta(h(n))$
B. $g(n) = O(f(n))$
C. $h(n) = \Omega(g(n))$
D. $g(n) = \Omega(h(n))$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider an array with the following elements

$$1\ 3\ 2\ 7\ 6\ 10\ 5\ 4.$$

The minimum number of interchanges required to make it a min-heap is ____

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #27** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Pick the correct order of asymptotic growth rate for the following functions.

$$n^{1/4}, e^n, 3^n, 1.001^n, n^{1000}, n^{\sqrt{n}}, n!$$

A. $n^{1/4} < n^{\sqrt{n}} < 1.001^n < e^n < 3^n < n! < n^{1000}$
B. $n^{1/4} < n^{1000} < 1.001^n < e^n < 3^n < n^{\sqrt{n}} < n!$
C. $1.001^n < n^{1/4} < n^{\sqrt{n}} < 1.001^n < e^n < 3^n < n^{1000} < n!$
D. $n^{1/4} < n^{1000} < n^{\sqrt{n}} < 1.001^n < e^n < 3^n < n!$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #28** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

The best data structures in terms of asymptotic time complexities for performing the following operations respectively are

- CONCATENATION (A, B): Concatenate all elements of A and B together
- Insert(T, key): Insert key to T
- Delete (T, key): Delete key from T if present
- Search (T, key): Returns true if key is present in T

A. Stack, Binary Search Tree, Sorted Array, Linked List
B. Linked List, Stack, Binary Search Tree, Sorted Array
C. Linked List, Stack, Sorted Array, Binary Search Tree
D. Stack, Linked List, Sorted Array, Binary Search Tree

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #29** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a STACK being implemented using two QUEUES by favoring the PUSH operations to POP. If $k$ PUSH operations and $l$ POP operations are done on this STACK with $n$ elements initially $(k, l << n)$, the minimum sum of the corresponding ENQUEUE and DEQUEUE operations will be

A. $O(kl)$
B. $O(ln)$
C. $O(k+l)$
D. $O(k)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #30** Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider the following two structure declarations and their initialization in C language.

```
struct node1
{
    int a1, a2;
}mynode1[10000];

for(i = 0; i < 10000; i++)
{
    mynode1[i].a1 = rand();
    mynode1[i].a2 = rand();
}
struct node2
{
    int a1[10000], a2[10000];
}mynode2;

for(i = 0; i < 10000; i++)
{
    mynode2.a1[i] = rand();
    mynode2.a2[i] = rand();
}
```

Assume size of integer as $4$ bytes, a $64$ byte cache line, and a $32KB$ fully associative cache implementation with all lines invalidated initially. If $n_1$ and $n_2$ denote the number of cache misses during the initializations of $mynode1$ and $mynode2$ respectively and if both $mynode1$ and $mynode2$ are aligned to cache lines with no other intervening cache accesses, the absolute value of $(n_1 - n_2) =$ ____

**Your Answer:** | **Correct Answer: 0** | **Not Attempted** | **Time taken: 00min 00sec** | **Discuss**

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